

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285623

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

CHO; Hanjin et al.

SPEECH RECOGNITION DEVICE AND OPERATING METHOD THEREOF

Abstract

The present disclosure is to improve noise in an always ready (AR) mode of a speech recognition device. The speech recognition device, according to embodiments of the present disclosure, may comprise: a main board; and a power board that under a ready mode in which switching elements are periodically operated, receives, from the main board, a signal for activating an AR mode for being always ready for speech recognition, supplies a certain voltage to the main board according to the received signal, and constantly operates the plurality of switching elements.

Inventors:	CHO; Hanjin (Seoul, KR), SIM; Jaedoo (Seoul, KR), KIM; Kyungmin (Seoul, KR)
Applicant:	LG ELECTRONICS INC. (Seoul, KR)
Family ID:	90191381
Assignee:	LG ELECTRONICS INC. (Seoul, KR)
Appl. No.:	19/105001
Filed (or PCT Filed):	September 08, 2022
PCT No.:	PCT/KR2022/013530

Publication Classification

Int. Cl.: G10L15/28 (20130101); G06F1/26 (20060101)

U.S. Cl.:

CPC G10L15/28 (20130101); G06F1/263 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a speech recognition device capable of recognizing speech.

BACKGROUND ART

[0002] Recently, the need for implementing speech recognition functions has increased with the use of artificial intelligence (AI) speakers or the like.

[0003] An Always Ready (AR) mode is a mode where speech recognition is always on standby.

[0004] Unlike the existing stand-by mode, the AR mode consumes more power than the existing stand-by mode because some circuits on the main board must be running at all times.

[0005] Conventional power boards perform burst operation to reduce power consumption of the power circuit in the standby mode.

[0006] In the case of burst operation, switching and resting periods occur alternately, causing the voltage and current ripples to increase.

[0007] Although there is a problem of noise generation in coils, switches, and multi-layer ceramic capacitors (MLCCs), in the normal standby mode, the current consumption is very low, less than 0.3 W or less and thus the noise is very low, so it does not cause problems in the field.

[0008] However, in the AR mode, the burst operation is the same as in standby mode, but power consumption increases, so noise problems due to voltage ripple and switching current occur.

[0009] At this time, it is difficult to improve noise with a simple circuit change, and there is a problem that the cost increases significantly if an electrolytic capacitor is added to reduce voltage ripple.

DISCLOSURE

Technical Problem

[0010] An object of the present disclosure is to improve noise caused by burst operation of a power board in an Always Ready mode for speech recognition.

[0011] An object of the present disclosure is to improve the noise caused by burst operation of a power board in an Always Ready mode without significantly increasing the cost of components.

Technical Solution

[0012] A speech recognition device according to an embodiment of the present disclosure may include a main board; and a power board receiving a signal from the main board to activate an Always Ready (AR) mode that always stands by for speech recognition, under a standby mode that periodically operates switching elements, and supplying a constant voltage to the main board according to the received signal, thereby continuously operating a plurality of switching elements.

[0013] An operating method of a speech recognition device including a main board and a power board according to an embodiment of the present disclosure may include operating the power board in a standby mode that periodically operates switching elements; under the standby mode, receiving a signal from the main board to activate an Always Ready (AR) mode that always stands by for speech recognition; and continuously operating a plurality of switching elements while supplying a constant voltage to the main board according to the received signal.

Advantageous Effect

[0014] According to various embodiments of the present disclosure, noise generation in an Always Ready mode for speech recognition can be significantly reduced.

[0015] According to various embodiments of the present disclosure, noise generation can be significantly reduced with little increase in cost by using simple circuits in an Always Ready mode for speech recognition.

Description

DESCRIPTION OF DRAWINGS

[0016] FIG. 1 is a view illustrating the configuration of a speech recognition device according to an embodiment of the present disclosure.

[0017] FIG. 2 is a block diagram illustrating the configuration of a main board according to one embodiment of the present disclosure.

[0018] FIG. 3 is a block diagram illustrating the configuration of a power board according to an embodiment of the present disclosure.

[0019] FIG. 4 is a flowchart for explaining an operating method of a speech recognition device according to an embodiment of the present disclosure.

[0020] FIG. 5 is a pin table including information on a plurality of pins provided in a power supply for connection to a main board according to one embodiment of the present disclosure.

[0021] FIG. 6 is a view explaining the configuration of an AR mode switching circuit according to one embodiment of the present disclosure.

[0022] FIG. 7 is a view explaining the voltage supplied to the main board and the operating waveform of the switching element under a normal mode according to the prior art.

[0023] FIG. 8 is a view explaining the voltage supplied to the main board and the operating waveform of the switching element under a standby mode according to the prior art.

[0024] FIG. 9 is a view explaining the voltage supplied to the main board and the operating waveform of the switching element under standby mode according to the prior art.

[0025] FIG. 10 is a view explaining the voltage supplied to the main board and the operating waveform of the switching element when the AR mode on signal is detected in the standby mode according to the embodiment of the present disclosure.

BEST MODE

[0026] Hereinafter, embodiments related to the present disclosure will be described in more detail with reference to the drawings. The suffixes “module” and “part” used for components in the following description are given or used interchangeably only for the convenience of writing the specification, and do not have distinct meanings or roles in themselves.

[0027] FIG. 1 is a view illustrating the configuration of a speech recognition device according to an embodiment of the present disclosure.

[0028] The speech recognition device **10** according to an embodiment of the present disclosure may be any one of a TV, an artificial intelligence speaker, a mobile terminal such as a smartphone/smart pad, but any electronic device capable of recognizing speech may be used.

[0029] Referring to FIG. 1, a speech recognition device **10** according to an embodiment of the present disclosure may include a main board **100** and a power board **200**.

[0030] The main board **100** can receive power from the power board **200**. According to the power mode of the power board **200**, only some components or all components of the main board **100** can be activated.

[0031] The main board **100** can be connected to the power board via a wired cable.

[0032] The power board **200** can supply voltage or current through a cable connected to the main board **100**.

[0033] The power board **200** can supply a voltage determined according to the power mode to the main board **100** and perform a switching operation according to the power mode.

[0034] Below, the power mode of the power board **200** may include a normal mode, a stand-by mode, and an always ready (AR) mode.

[0035] The normal mode may be a mode that continuously operates the switching elements included in the switching part **230** of the power board **200**. Under the normal mode, the power board **200** may provide a first voltage to the main board **100**. The first voltage may be 12V, but this

is only an example.

[0036] Under the normal mode, burst operations are not performed and switching operations can be performed continuously.

[0037] The stand-by mode is a mode in which speech recognition is periodically put on standby and may be a mode in which power is periodically supplied to some components of the main board **100**.

[0038] In standby mode, the power board **200** can operate the switching elements of the switching part **230** alternately between switching and resting periods to reduce power consumption.

[0039] Under standby mode, the power board **200** can supply a second voltage lower than the first voltage to the main board **100** to reduce power consumption. The first voltage may be 7.8 V, but this is only an example.

[0040] The operation of switching elements that alternate between switching and resting periods periodically can be referred to as a burst operation.

[0041] In the standby mode, power can be supplied only to the minimum components for controlling the power of the speech recognition device **10**.

[0042] An Always Ready (AR) mode is a mode in which speech recognition is always on standby, and may be a mode in which power is always supplied to some components of the main board **100**.

[0043] In the Always Ready mode, the power board **200** can supply a second voltage lower than the first voltage to the main board **100**.

[0044] In the Always Ready mode, the power board **200** can continuously operate the switching elements included in the switching part **230** while supplying the first voltage to the main board **100**.

[0045] In other words, the constant standby mode and normal mode may have the same switching operation of the switching elements, but there is only a difference in the magnitude of the voltage supplied to the main board **100**.

[0046] For example, the Always Ready mode may be a mode in which power is always supplied to the microphone, audio processor, and microcomputer included in the main board **100** to always recognize speech.

[0047] Each component of the main board **100** and power board **200** may be provided in the form of an IC (Integrated Chip).

[0048] FIG. **2** is a block diagram illustrating the configuration of a main board according to one embodiment of the present disclosure.

[0049] Referring to FIG. **2**, a main board **100** according to an embodiment of the present disclosure may include a power interface **110**, a DC/DC converter **120**, an external input interface **130**, a user input interface **140**, a microphone **150**, an audio processor **160**, a speaker **170**, and a microcomputer **190**.

[0050] Each component of the main board **100** may be provided in the form of an IC (Integrated Chip).

[0051] The power interface **110** may be an interface for receiving power from the power board **200**.

[0052] The power interface **110** can receive voltage and current from the power board **200** through a cable.

[0053] The DC/DC converter **120** can regulate the voltage supplied from the power board **200** and supply the regulated voltage to each component of the main board **100**.

[0054] A plurality of DC/DC converters **120** may be provided.

[0055] The external input interface **130** may be an interface for connection with an external device. The external input interface **130** may include one or more HDMI (High Definition Multimedia Interface) terminals and one or more USB (Universal Serial Bus) terminals.

[0056] The user input interface **140** can receive commands for controlling the speech recognition device **10** from a remote control.

[0057] The user input interface **140** may include one or more of an infrared module or a Bluetooth module.

[0058] The microphone **150** can receive a speech signal from outside.

[0059] The audio processor **160** can preprocess a speech signal received from outside.

[0060] The speaker **170** can output a response speech signal based on a speech signal.

[0061] The microcomputer **190** can control the overall operation of the speech recognition device **10**.

[0062] The microcomputer **190** can generate an Always Ready (AR) mode on signal.

[0063] The microcomputer **190** can transmit an AR mode on signal to the power board **200** through the power interface **110**.

[0064] The main board **100** may further include a short-range communication module, such as a WiFi module.

[0065] The main board **100** may further include a timing controller for driving a display panel connected to the main board **100**.

[0066] FIG. **3** is a block diagram illustrating the configuration of a power board according to an embodiment of the present disclosure.

[0067] Referring to FIG. **3**, the power board **200** may include a power supply **210**, an AR mode switching circuit **220**, a switching part **230**, and a controller **290**.

[0068] The power supply **210** may be an interface for supplying power to the main board **100**. The power supply **210** may include 24 pins. Any one of the 24 pins (AR mode input pin) may be for receiving an AR mode on signal.

[0069] The AR mode switching circuit **220** can convert an AR mode on signal received through an AR mode input pin into an optical output signal and transmit the converted optical output signal to the controller **290**.

[0070] The controller **290** can detect the AR mode based on the received optical output signal.

[0071] The switching part **230** may include a plurality of switching elements. Each of the plurality of switching elements may be a transistor. The plurality of switching elements may generate current through a switching operation and supply the generated current to the main board **100**.

[0072] The controller **290** can control the overall operation of the power board **200**.

[0073] The controller **290** can check whether an AR mode on signal is detected through the AR mode input pin **211** under standby mode.

[0074] The controller **290** can continuously operate the switching elements at the voltage level of the standby mode when the AR mode on signal is detected.

[0075] The controller **290** can generate a voltage level supplied to the main board **100** in a flat form in standby mode and provide the generated flat voltage to the main board **100**.

[0076] For this purpose, the controller **290** may further include a rectifier circuit that converts the voltage of the triangle wave into a flat form.

[0077] The controller **290** can control the switching part **230** so that the switching elements maintain switching operation while supplying a flat voltage to the main board **100**.

[0078] FIG. **4** is a flowchart for explaining an operating method of a speech recognition device according to an embodiment of the present disclosure.

[0079] The microcomputer **190** of the main board **100** generates an Always Ready (AR) mode on signal (**S401**).

[0080] In one embodiment, the microcomputer **190** can receive a command to operate in AR mode via the user input interface **140**. For example, the user input interface **140** can receive a command to activate AR mode via a remote control.

[0081] The microcomputer **190** can generate an AR mode on signal upon receiving a command to operate in AR mode.

[0082] The AR mode on signal may be a signal to switch the power mode of the power board **200** to AR mode.

[0083] The microcomputer **190** of the main board **100** transmits an AR mode on signal to the power board **200** through the power interface **110** (**S403**).

[0084] The power interface **110** may include a cable connecting the main board **100** and the power board **200**, and may transmit an AR mode on signal to the power supply **210** of the power board **200** through the cable.

[0085] The controller **290** of the power board **200** checks whether an AR mode on signal is detected under standby mode (S405).

[0086] The controller **290** may be operating in standby mode and may determine whether an AR mode on signal is detected through the power supply **210** under the standby mode.

[0087] The power supply **210** may include a plurality of pins for connection to the power interface **110** of the main board **100**.

[0088] Among the plurality of pins, the reserved pin can be used to receive the AR mode on signal.

[0089] This is explained with reference to FIG. 5.

[0090] FIG. 5 is a pin table including information on a plurality of pins provided in a power supply for connection to a main board according to one embodiment of the present disclosure.

[0091] The pin table **500** may include information about 24 pins connecting the main board **100** and the power board **200**.

[0092] The pin table **500** can include 24 pins, and each pin can display information about the applied driving voltage and information about the ground.

[0093] Among the 24 pins, pin 1 may be a pin to which no function is applied for future use.

[0094] In an embodiment of the present disclosure, pin 1 may be used to receive an AR mode on signal from the main board **100**.

[0095] In other words, pin 1 can be used as a pin to receive the AR mode on signal. Again, FIG. 4 is explained.

[0096] When the controller **290** of the power board **200** detects an AR mode on signal, it continuously operates the switching elements at the voltage level of the standby mode (S407).

[0097] In one embodiment, the controller **290** can determine whether an AR mode on signal has been received through the AR mode input pin **211** provided in the power supply **210**.

[0098] When operation in AR mode is detected, the controller **290** can continuously operate the switching elements included in the switching part **230** while supplying the voltage level of the standby mode to the main board **100**.

[0099] The controller **290** of the power board **200** maintains the power mode in the standby mode when the AR mode on signal is not detected (S409).

[0100] FIG. 6 is a view explaining the configuration of an AR mode switching circuit according to one embodiment of the present disclosure.

[0101] The AR mode switching circuit **220** of FIG. 6 may include a filter **221**, a bipolar transistor **222**, and a photocoupler **223**.

[0102] The filter **221** can filter the AR mode on signal input from the main board **100** through the AR mode input pin **211**.

[0103] The filter **221** may include one or more resistors and capacitors.

[0104] The bipolar transistor **222** can be turned on according to the AR mode on signal filtered by the filter **221**.

[0105] A photocoupler **223** can transmit input electrical signals and output electrical signals as light by inserting an insulating material between a light-emitting element and a light-receiving element.

[0106] A photocoupler **223** can be used as an interface to transmit signals between circuits with different ground potentials.

[0107] When the controller **290** directly receives the AR mode on signal from the main board **100**, the main board **100** and the power board **200** use different types of grounding, so excessive current may flow through the controller **290**.

[0108] To prevent this, the photocoupler **223** can output an AR mode on signal or an AR mode off signal in the form of light.

[0109] The controller **290** can operate the power mode of the speech recognition device **10** in AR

mode by detecting an AR mode on signal in the form of light.

[0110] The controller **290** can operate the power mode of the power board **200** in AR mode when the optical output signal of the photo coupler **223** indicates an AR mode on signal.

[0111] The controller **290** can operate the switching part **230** as in normal mode while supplying a second voltage to the main board **100** that is lower than the first voltage provided to the main board **100** under normal mode.

[0112] The AR mode switching circuit **220** illustrated in FIG. **6** is merely an example, and may have a circuit configuration in which an AR mode on signal is transmitted in the form of a magnetic field.

[0113] In other words, the AR mode switching circuit **220** may convert the AR mode on signal into an optical output signal through a photocoupler **223**, or may convert the AR mode on signal into a magnetic field output signal through a transformer (not illustrated).

[0114] Next, the power modes according to the prior art and the AR mode according to the embodiment of the present disclosure are compared and explained.

[0115] FIG. **7** is a view explaining the voltage supplied to the main board and the operating waveform of the switching element under a normal mode according to the prior art.

[0116] Referring to FIG. **7**, waveforms of voltage and current supplied from the power board **200** to the main board **100** under the normal mode are illustrated.

[0117] The voltage waveform **710** is a 12V voltage in a flat form. The DC/DC converter **120** of the main board **100** can step down the voltage to 12V and supply it to each component of the main board **100**.

[0118] The current waveform **730** continuously displays a peak waveform due to the continuous switching operation of the switching elements provided in the switching part **230**.

[0119] In the normal mode, power consumption can range from tens to hundreds of watts.

[0120] FIG. **8** is a view explaining the voltage supplied to the main board and the operating waveform of the switching element under a standby mode according to the prior art.

[0121] Referring to FIG. **8**, waveforms of voltage and current supplied from the power board **200** to the main board **100** under the standby mode are illustrated.

[0122] The voltage waveform **810** has a triangular wave or sawtooth wave shape with an average voltage of 7.8V. When the power board **200** is in standby mode, it can supply 7.8V, which is less than the 12V supplied in normal mode, to the main board **100**.

[0123] The current waveform **830** is a peak waveform that appears periodically due to repeated operation (burst operation) of periodic switching operation and resting operation of the switching elements provided in the switching part **230**.

[0124] FIG. **9** is a view explaining the voltage supplied to the main board and the operating waveform of the switching element under standby mode according to the prior art.

[0125] Referring to FIG. **9**, waveforms of voltage and current supplied from the power board **200** to the main board **100** under the conventional Always Ready mode are illustrated.

[0126] The voltage waveform **910** has a triangular wave or sawtooth wave shape with an average voltage of 7.8V. The voltage supplied to the main board **100** swings around 7 to 10V, and the voltage waveform **910** has a ripple component.

[0127] The current waveform **930** is a peak waveform that appears periodically due to repeated operation (burst operation) of periodic switching operation and resting operation of the switching elements provided in the switching part **230**.

[0128] What is different from the standby mode is that the burst operation is performed, but the switching interval is longer so that power is continuously supplied to some components of the main board **100**, so that the peak interval of the current waveform **930** increases.

[0129] Accordingly, in the conventional AR mode, there was a problem in that noise was generated significantly by the coils, switching elements, and MLCCs provided in the power board **200** due to voltage ripple and excessive current peaks.

[0130] FIG. 10 is a view explaining the voltage supplied to the main board and the operating waveform of the switching element when the AR mode on signal is detected in the standby mode according to the embodiment of the present disclosure.

[0131] The power board **200** can supply a voltage of 7.8V in a flat form to the main board **100** based on the AR mode on signal received from the main board **100**. At the same time, the power board **200** can continuously operate the switching elements included in the switching part **230** similar to the switching operation of the normal mode, instead of the burst operation.

[0132] Accordingly, the voltage waveform **1010** may have a flat shape of 7.8 V, and the current waveform **1030** may have a shape with a reduced peak waveform.

[0133] In other words, when the power board **200** detects the AR mode through the AR mode input pin **211**, the magnitude of the voltage supplied to the main board **100** is set to the level of the standby mode to produce a flat waveform, and the current peak waveform can be reduced by continuing the switching operation of the switching elements.

[0134] Accordingly, the noise of the components provided on the power board **200** can be reduced.

[0135] According to an embodiment of the present disclosure, noise improvement in AR mode can be achieved dramatically with almost no increase in cost by using an AR mode switching circuit **220**.

[0136] The peak noise level measured under AR mode according to the prior art is 25 dB, but the peak noise level measured under AR mode according to the embodiment of the present disclosure is significantly improved to 18.4 dB.

[0137] According to one embodiment of the present disclosure, the method described above can be implemented as a processor-readable code on a medium in which a program is recorded. Examples of the processor-readable medium include a ROM, a RAM, a CD-ROM, a magnetic tape, a floppy disk, an optical data storage device, and the like.

[0138] The display device described above is not limited to the configuration and method of the embodiments described above, and the embodiments may be configured by selectively combining all or part of the embodiments so that various modifications can be made.

Claims

1. A speech recognition device comprising: a main board; and a power board receiving a signal from the main board to activate an Always Ready (AR) mode that always stands by for speech recognition, under a standby mode that periodically operates switching elements, and supplying a constant voltage to the main board according to the received signal, thereby continuously operating a plurality of switching elements.
2. The speech recognition device of claim 1, wherein the power board includes an AR mode input pin for receiving a signal for activating the AR mode.
3. The speech recognition device of claim 2, wherein the power board further includes a photocoupler for converting a signal received through the AR mode input pin into an optical output signal, and detects operation in the AR mode based on the optical output signal received through the photocoupler.
4. The speech recognition device of claim 1, wherein the constant voltage is the voltage level supplied to the main board in the standby mode, and wherein the waveform of the constant voltage is a flat form.
5. The speech recognition device of claim 1, wherein the main board receives a command for switching to the AR mode through user input and generates the signal for activating the AR mode according to the received command.
6. The speech recognition device of claim 1, wherein the power board includes: a switching part including the plurality of switching elements, a power supply, which is an interface that supplies power to the main board, an AR mode switching circuit converting the signal received from the

main board; and a controller switching the standby mode of the speech recognition device to the AR mode based on the converted signal.

7. The speech recognition device of claim 1, wherein the power mode of the speech recognition device includes the standby mode, the AR mode, and a normal mode, wherein the magnitude of the constant voltage supplied to the main board under the AR mode is the same as the magnitude of the voltage supplied to the main board in the standby mode, and wherein operations of the plurality of switching elements under the AR mode is the same as the switching operation in the normal mode.

8. The speech recognition device of claim 1, wherein the speech recognition device is either a TV, an AI speaker, or a smartphone.

9. An operating method of a speech recognition device including a main board and a power board comprising: operating the power board in a standby mode that periodically operates switching elements; under the standby mode, receiving a signal from the main board to activate an Always Ready (AR) mode that always stands by for speech recognition; and continuously operating a plurality of switching elements while supplying a constant voltage to the main board according to the received signal.

10. The operating method of a speech recognition device of claim 9, wherein the power board includes an AR mode input pin for receiving a signal for activating the AR mode.

11. The operating method of a speech recognition device of claim 10, wherein the power board further includes a photocoupler for converting a signal received through the AR mode input pin into an optical output signal, and detects operation in the AR mode based on the optical output signal received through the photocoupler.

12. The operating method of a speech recognition device of claim 9, wherein the constant voltage is the voltage level supplied to the main board in the standby mode, and wherein the waveform of the constant voltage is a flat form.

13. The operating method of a speech recognition device of claim 9, further comprising: receiving a command for switching to the AR mode through user input by the main board; and generating the signal for activating the AR mode according to the received command.

14. The operating method of a speech recognition device of claim 9, further comprising: converting the signal received from the main board into an optical output signal or a magnetic field output signal; and switching the standby mode of the speech recognition device to the AR mode based on the converted signal.

15. The operating method of a speech recognition device of claim 9, wherein the power mode of the speech recognition device includes the standby mode, the AR mode, and a normal mode, wherein the magnitude of the constant voltage supplied to the main board under the AR mode is the same as the magnitude of the voltage supplied to the main board in the standby mode, and wherein operations of the plurality of switching elements under the AR mode is the same as the switching operation in the normal mode.
